

ROBIEI, Switzerland

WMO: 067510

Lat: 46.4431N

Lon: 8.5133E

Elev: 1891

StdP: 80.58

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.3	-11.6	-28.7	0.3	-6.9	-25.5	0.5	-6.1	11.4	-4.9	10.1	-4.3	3.3	350	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	19.6	12.2	18.2	11.8	16.9	11.3	14.0	17.0	13.1	16.2	12.3	15.4	2.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	11.6	15.4	11.8	10.9	14.5	11.0	10.3	13.7	45.8	17.1	43.1	16.2	40.7	15.4	17.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.4	6.2	DB	-14.9	21.9	2.5	1.7	-16.7	23.1	-18.2	24.1	-19.5	25.0	-21.3	26.2	(4)
(5)			WB	-16.3	15.5	2.1	1.1	-17.9	16.3	-19.1	17.0	-20.3	17.6	-21.9	18.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-3.4	-3.6	-1.2	1.8	5.2	10.5	12.5	12.0	8.5	5.1	0.5	-2.3	(6)
(7)		DBStd	6.80	4.11	4.45	3.99	3.34	3.30	3.42	2.77	2.91	3.14	3.68	3.90	4.12	(7)
(8)		HDD10.0	2485	416	382	347	245	152	34	10	13	63	156	287	380	(8)
(9)		HDD18.3	5289	674	616	606	494	406	236	181	197	294	411	537	638	(9)
(10)		CDD10.0	239	0	0	0	0	4	49	88	74	19	3	1	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	1	0	1	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	1	0	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.4	2.3	2.3	2.0	2.1	2.0	1.7	1.6	1.6	1.5	1.9	2.2	(14)
(15)	Precipitation	PrecAvg	1824	87	85	95	162	192	179	148	177	201	192	211	93	(15)
(16)		PrecMax	2520	262	283	303	453	554	414	359	314	612	693	755	233	(16)
(17)		PrecMin	1011	18	9	7	20	44	48	47	49	13	9	17	4	(17)
(18)		PrecStd	379	58	60	67	115	118	90	65	60	155	145	184	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.1	8.7	11.6	15.6	21.2	21.0	21.7	17.9	14.9	12.3	7.9	(19)
(20)			MCWB	0.8	1.6	1.1	4.6	8.9	11.8	13.3	12.1	10.0	6.6	4.1	0.6	(20)
(21)		2%	DB	6.1	7.0	7.2	9.3	13.3	19.0	19.3	19.4	15.7	13.0	9.3	6.3	(21)
(22)			MCWB	0.0	0.4	0.8	3.7	7.1	11.4	12.2	12.4	9.8	5.9	3.1	0.1	(22)
(23)		5%	DB	4.5	4.6	5.9	7.8	11.5	17.4	18.0	17.8	14.3	11.3	7.4	4.8	(23)
(24)			MCWB	-0.6	-0.8	0.3	2.8	6.3	11.0	12.0	12.0	9.1	5.3	2.1	-0.7	(24)
(25)		10%	DB	2.6	2.7	4.5	6.6	10.0	15.8	16.7	16.3	13.0	9.8	5.6	3.4	(25)
(26)			MCWB	-1.9	-1.9	-0.2	2.3	5.4	10.3	11.4	11.3	8.6	5.0	1.1	-1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.5	2.7	5.8	10.0	14.1	15.6	15.2	12.5	9.1	5.7	2.3	(27)
(28)			MCDWB	5.8	7.4	6.3	9.2	14.2	17.4	18.5	18.5	14.8	11.6	8.9	5.8	(28)
(29)		2%	WB	0.9	0.9	1.8	4.5	8.2	13.0	14.1	14.0	11.4	8.0	4.0	1.1	(29)
(30)			MCDWB	4.9	6.0	5.0	7.6	11.7	16.6	17.0	16.9	13.8	9.5	7.9	4.1	(30)
(31)		5%	WB	-0.3	-0.5	1.1	3.7	7.1	11.9	13.2	13.0	10.4	7.3	2.9	0.1	(31)
(32)			MCDWB	3.3	3.8	4.2	6.7	10.2	16.1	16.3	15.9	12.8	9.0	5.5	2.8	(32)
(33)		10%	WB	-1.5	-1.6	0.4	2.9	6.2	10.9	12.3	12.2	9.5	6.5	2.1	-0.7	(33)
(34)			MCDWB	1.7	1.7	3.4	5.8	9.0	14.9	15.5	14.9	11.9	8.3	4.4	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.2	5.1	4.9	4.9	5.6	5.7	5.2	5.1	4.8	4.3	4.8	(35)
(36)			MCDBR	6.7	7.0	6.3	6.2	7.0	7.6	7.2	6.9	6.5	6.8	6.3	6.2	(36)
(37)			MCWBR	4.5	4.1	3.9	3.8	4.2	4.8	4.7	4.6	4.5	4.1	3.7	4.0	(37)
(38)		5% WB	MCDBR	6.3	6.1	5.2	5.2	5.8	6.7	6.4	5.7	5.6	4.8	4.7	5.2	(38)
(39)			MCWBR	4.7	4.3	3.6	3.7	4.1	4.9	4.8	4.5	4.4	3.7	3.7	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.232	0.257	0.299	0.344	0.361	0.370	0.366	0.356	0.333	0.307	0.265	0.231	(40)	
(41)		taud	2.333	2.279	2.222	2.204	2.273	2.295	2.303	2.345	2.401	2.453	2.447	2.369	(41)	
(42)		Ebn at Noon	889	927	926	909	903	894	893	888	881	858	841	852	(42)	
(43)		Edh at Noon	72	95	118	133	129	127	125	115	99	80	65	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.61	2.57	3.79	4.47	4.81	5.66	5.66	4.91	3.82	2.52	1.56	1.34	(44)	
(45)		RadStd	0.20	0.30	0.55	0.72	0.48	0.49	0.48	0.43	0.37	0.30	0.24	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	N/A		
(47) Regional (121 neighbors)		N/A	N/A	N/A	+0.73	+0.42	+0.32	N/A	N/A	+57	+15		

Nomenclature: See separate page